PRAVEENA KIRAN

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Eligible to work in Japan (Permanent Resident) · JLPT N3

SUMMARY

Computer Science student (B.Tech, graduating 2028) who builds and ships real software: an LLM engine that diagnoses and patches failing code, a released gesture- and voice-controlled accessibility app, and full-stack web apps. Strong in Python and Java, object-oriented design, automated testing, and debugging complex systems. Works comfortably across cultures in English and Japanese (JLPT N3).

EDUCATION

Bachelor of Technology in Computer Science and Engineering — Woxsen University, Telangana, India — 2024 – 2028 (expected)
GPA 8.7 / 10

PROJECTS

Phoenix — Self-Healing Software System · Python · LLMs (Groq API) · Pytest — Team project · 2026

- Built the LLM engine (~4,400 lines incl. tests) that diagnoses the root cause of production incidents from logs and evidence and generates multi-file code patches.
- Added sandbox validation of every patch, fallback patches and API error handling so failures degrade safely instead of breaking the system.
- Wrote an evaluation harness and Pytest suites to measure diagnosis and patch reliability before and after prompt tuning.

Wield — Hands-Free Desktop Control (Gestures + Voice) · Python · OpenCV · MediaPipe · PySide6 · Vosk — Lead developer · 2026
Live: praveena-kiran.github.io/Wield · Code: github.com/Praveena-Kiran/Wield

- Shipped v1.0.0 of a Windows accessibility app that controls the computer and PowerPoint hands-free with hand gestures, drawn motions and offline voice commands (“Hey Wield, next slide”).
- Built real-time hand tracking with confidence-based recognition, gesture stabilization and a confirm gesture to cut false triggers; users record their own gestures and motions, matched with DTW over fingertip or palm trajectories.
- Made every action configurable as any mix of gesture, motion and voice, with sound cues and an on-screen overlay; camera and mic stay off until a session starts, and speech is processed on-device.
- Wrote 172 automated tests and GitHub Actions CI that builds a standalone PyInstaller app and runs a packaged self-test before each release.

Found It — Campus Lost & Found Platform · React · Flask · MongoDB · Google Cloud Vision — Hackathon team · 2025

- Built the Flask REST backend, data models and JWT authentication (bcrypt), plus the React login, sign-up, report and browse pages.
- Integrated Google Cloud Vision so photos of lost items are matched to found-item reports instead of manual text search.

LinuxMonitor — Bash System Monitor · Bash · Linux · WSL2 · Git — Solo · 2026

- Live terminal monitor for CPU, memory, disk, load, processes and connectivity with NORMAL / WARNING / CRITICAL thresholds and service health checks.

SKILLS

Languages	Python, Java, JavaScript, SQL, Bash
Engineering	Object-oriented design, REST APIs, automated testing (Pytest, unittest), CI/CD (GitHub Actions), debugging, Git / GitHub
Web & data	React, Flask, HTML/CSS, MySQL, MongoDB
AI, vision & speech	LLM APIs (Groq), prompt evaluation, OpenCV, MediaPipe, Vosk (offline speech), Google Cloud Vision
Desktop	PySide6 / Qt, PyInstaller packaging
Systems & cloud	Linux, WSL2, AWS fundamentals

LEADERSHIP

President, Woxsen Japan Centre — May 2025 – Present

- Oversee and coordinate activities for a 150+ member student organization.
- Coordinated a 120-hour JLPT N5 program with NaRa Japan Hub for 90+ students.
- Coordinate engagement with 15+ delegations from Japanese universities, organizations and industry partners; helped deliver Kizuna Matsuri with The Japan Foundation and Toshiba T&D (India).

CERTIFICATIONS & LANGUAGES

Object-Oriented Design — University of Alberta · Java Programming: Solving Problems with Software — Duke University · Introduction to Cloud Computing on AWS — Udemy · JLPT N3

English (professional) · Japanese (conversational, JLPT N3) · Hindi (professional) · Telugu (native)